



Rally Detection V2 - Complete Setup & Usage Guide



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What's New in V2

Critical Enhancements (Phase 1)

- ✓ **Height-based Ground Detection** - Stops tracking when shuttlecock hits ground
- ✓ **Speed-based Activity Validation** - Filters out slow-moving objects
- ✓ **Inter-rally Gap Enforcement** - Auto-merges rallies with short gaps

Quality Improvements (Phase 2)

- ✓ **Trajectory Smoothness** - Rejects erratic position jumps
- ✓ **ROI/Boundary Filtering** - Only tracks within court boundaries

Optional Enhancements (Phase 3)

- ✓ **Confidence Trend Analysis** - Validates detection quality over time

Expected Improvements

- **60-70%** error reduction from Phase 1
- **15-20%** additional improvement from Phase 2

- **5-10%** additional improvement from Phase 3
 - **Overall: ~80-100%** improvement over V1
-

Installation

Basic Installation (Color Detection Only)



bash

```
# Install basic dependencies
```

```
pip install opencv-python numpy pandas
```

```
# You're ready to go!
```

```
python rally_detector_v2.py --video sample.mp4
```

Option 1: TrackNetV3 (Recommended - Best Accuracy)

Step 1: Clone TrackNetV3



bash

```
git clone https://github.com/alenzenx/TrackNetv3
```

```
cd TrackNetv3
```

Step 2: Install Dependencies



bash

```
pip install torch torchvision
pip install opencv-python numpy pandas
```

Step 3: Download/Train Model



bash

```
# Option A: Download pre-trained weights (if available)
# Check the repository for weight files

# Option B: Train your own model
# Follow TrackNetV3 training instructions with your badminton footage
python train.py --data ./data --epochs 50
```

Step 4: Copy V2 Script



bash

```
# Place rally_detector_v2.py in the project directory
cd ..
# Make sure TrackNetV3 folder is in same directory as script
```

Option 2: Monotrack



bash

Coming soon - integration pending
Will follow similar setup process

Option 3: YOLO-based Detection



bash

pip install ultralytics
pip install opencv-python numpy pandas

Option A: Use YOLOv8n (generic)
Will use sports ball class - less accurate but works out of box

Option B: Train custom shuttlecock detector
Use Roboflow or train with your dataset

Quick Start

For Your sample.mp4 File

Method 1: Simplest (Color Detection)



bash

```
python rally_detector_v2.py --video sample.mp4
```

Method 2: With Visualization



bash

```
python rally_detector_v2.py --video sample.mp4 --visualize
```

Method 3: Save Annotated Video



bash

```
python rally_detector_v2.py --video sample.mp4 --visualize \  
--save-video sample_annotated.mp4
```

Method 4: With TrackNetV3 (Best Results)



bash

```
python rally_detector_v2.py --video sample.mp4 --visualize \  
--model tracknet \  
--model-path ./TrackNetV3/weights/best.pt
```

Model Options

1. Color Detection (Default)

Pros: No setup required, works immediately

Cons: Less accurate, struggles with lighting changes

Use when: Testing, quick results needed



bash

```
python rally_detector_v2.py --video sample.mp4 --model color
```

2. TrackNetV3 (Recommended)

Pros: Specifically designed for ball/shuttlecock tracking

Cons: Requires setup and training/weights

Use when: Production use, best accuracy needed



bash

```
python rally_detector_v2.py --video sample.mp4 \  
  --model tracknet \  
  --model-path ./TrackNetV3/weights/best.pt
```

3. YOLO

Pros: Good general object detection, pre-trained models available

Cons: Not specialized for shuttlecock, needs fine-tuning

Use when: Have custom trained model or testing



bash

```
python rally_detector_v2.py --video sample.mp4 \
  --model yolo \
  --model-path ./weights/shuttlecock_yolo.pt
```

4. Monotrack

Pros: Good for tracking sports objects

Cons: Integration pending

Status: Coming soon



Running Examples

Example 1: Basic Run (Recommended Start)



bash

```
python rally_detector_v2.py --video sample.mp4 --visualize
```

Output:

- rallies_v2_output.csv - Rally timings
- rallies_v2_output_stats.json - Detection statistics
- Live visualization window

Example 2: Production Run with TrackNetV3



bash

```
python rally_detector_v2.py --video sample.mp4 \  
  --model tracknet \  
  --model-path ./TrackNetV3/weights/best.pt \  
  --output match1_rallies.csv \  
  --save-video match1_annotated.mp4 \  
  --confidence 0.5
```

Example 3: Adjust V2 Parameters



bash

```
python rally_detector_v2.py --video sample.mp4 \  
  --ground-ratio 0.7 \  
  --min-speed 40 \  
  --inter-gap 2.5 \  
  --max-jump 150 \  
  --min-duration 1.5
```

Example 4: Evaluation Against Ground Truth



bash


```
python rally_detector_v2.py --video sample.mp4 \  
--output predicted.csv \  
--ground-truth ground_truth.csv
```

Example 5: Disable ROI Filtering



bash

```
python rally_detector_v2.py --video sample.mp4 --no-roi
```

Parameter Tuning Guide

When to Adjust Parameters

Ground Ratio (--ground-ratio)

Default: 0.75 (bottom 25% is ground)

Adjust if:

- Rallies ending too early → **Increase** to 0.8
- Tracking continues on ground → **Decrease** to 0.7



bash

More strict (ends rally earlier)

--ground-ratio 0.7

More lenient (tracks lower)

--ground-ratio 0.8

Minimum Speed (--min-speed)

Default: 50 px/frame

Adjust if:

- Missing slow rallies → **Decrease** to 30-40
- Tracking stationary objects → **Increase** to 60-70



bash

Track slower movements

--min-speed 30

Only fast shuttlecock

--min-speed 70

Inter-Rally Gap (--inter-gap)

Default: 3.0 seconds

Adjust if:

- Split rallies should be merged → **Increase** to 4-5
- Different rallies being merged → **Decrease** to 2.0



bash

```
# Merge more aggressively
--inter-gap 4.5
```

```
# Less merging
--inter-gap 2.0
```

Max Position Jump (--max-jump)

Default: 200 pixels

Adjust if:

- Fast serves rejected → **Increase** to 250-300
- Erratic tracking continues → **Decrease** to 150



bash

```
# Allow larger jumps
--max-jump 300
```

```
# More strict smoothing
--max-jump 150
```

Minimum Duration (--min-duration)

Default: 2.0 seconds

Adjust if:

- Short rallies missed → **Decrease** to 1.0-1.5
- Too many false positives → **Increase** to 3.0



bash

Capture shorter rallies

`--min-duration 1.5`

Only long rallies

`--min-duration 3.0`

Understanding the Visualization

When running with `--visualize`, you'll see:

Color Coding

-  **Green circle** - Current shuttlecock position
-  **Yellow line** - Shuttlecock trajectory
-  **Blue rectangle** - ROI (valid court area)
-  **Red line** - Ground detection threshold

Status Display



V2 | State: RALLY_ACTIVE | Rally: 3

00:45

Speed: 65.3px/f

States

- **IDLE** - Waiting for rally to start (gray)
- **RALLY_ACTIVE** - Rally in progress (green)
- **RALLY_ENDING** - Confirming rally end (yellow)

 Output Files

1. CSV File (rallies_v2_output.csv)



csv

```
video_id,rally_id,start_time,end_time,duration
sample,1,00:12,00:28,16.0
sample,2,00:35,00:51,16.0
sample,3,01:05,01:22,17.0
```

2. Statistics File (rallies_v2_output_stats.json)



json

```
{  
  "total_rallies": 15,  
  "rejected_ground": 8,  
  "rejected_speed": 12,  
  "rejected_trajectory": 3,  
  "rejected_roi": 5,  
  "rallies_merged": 4  
}
```

3. Annotated Video (optional)

- MP4 file with visualization overlays
 - Shows detection in action
 - Useful for debugging
-

Troubleshooting

Problem: No rallies detected

Solutions:

1. Try color detection first to verify video
2. Lower confidence threshold: `--confidence 0.3`
3. Adjust ground ratio: `--ground-ratio 0.8`
4. Disable ROI: `--no-roi`
5. Check if shuttlecock is visible in video

Problem: Too many false positives

Solutions:

1. Increase confidence: `--confidence 0.6`
2. Increase min speed: `--min-speed 60`
3. Decrease max jump: `--max-jump 150`

4. Increase min duration: `--min-duration 3.0`

Problem: Rallies split incorrectly

Solutions:

1. Increase inter-gap: `--inter-gap 4.0`
2. Lower ground ratio: `--ground-ratio 0.7`
3. Check visualization to see why splits occur

Problem: TrackNetV3 import error

Solutions:

1. Verify TrackNetV3 folder in same directory as script
2. Check Python path
3. Install torch: `pip install torch torchvision`
4. Fallback to color detection temporarily

Problem: Video won't open

Solutions:

1. Check file path is correct
2. Try absolute path: `/full/path/to/sample.mp4`
3. Verify video codec compatibility
4. Test with VLC player first

Problem: Slow processing

Solutions:

1. Use color detection instead of neural network
 2. Don't use `--visualize` for faster processing
 3. Process shorter clips for testing
 4. Ensure GPU available for TrackNetV3
-



Evaluation

Creating Ground Truth

Create a CSV file with actual rally times:



csv

```
video_id,rally_id,start_time,end_time,duration
sample,1,00:15,00:32,17
sample,2,00:45,01:02,17
```

Run Evaluation



bash

```
python rally_detector_v2.py --video sample.mp4 \
  --output predicted.csv \
  --ground-truth ground_truth.csv
```

Metrics Explained

- **Precision:** % of detected rallies that are correct
 - **Recall:** % of actual rallies that were detected
 - **F1-Score:** Harmonic mean of precision and recall
 - **Average IoU:** Temporal overlap accuracy
-

Best Practices

1. **Start Simple:** Run with defaults first
 2. **Visualize:** Always use `--visualize` during testing
 3. **Iterate:** Adjust one parameter at a time
 4. **Document:** Save your best parameter combinations
 5. **Validate:** Compare against ground truth regularly
 6. **Model:** Invest time in TrackNetV3 for production use
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Next Steps

1. **Test on sample.mp4** with default settings
 2. **Review visualization** to understand behavior
 3. **Tune parameters** based on your video characteristics
 4. **Consider TrackNetV3** if accuracy insufficient
 5. **Create ground truth** for your videos
 6. **Evaluate** and iterate
-

Support

For issues or questions:

1. Check this guide first
2. Review console output for errors
3. Test with visualization enabled
4. Try different parameter combinations
5. Verify video quality and shuttlecock visibility

Good luck with your rally detection! 